

2 **Goodness-of-fit for multi-distribution neutrino** 3 **cross-section measurements with shared events**

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8 **ABSTRACT:** When multiple differential cross-section measurements are extracted from a common
9 event sample, the event-sharing structure between distributions imposes exact linear constraints
10 among the bin counts that reduce the effective dimensionality of the measurement below the total bin
11 count. When the statistical covariance matrix accounts for these inter-distribution correlations, the
12 constraints manifest as a rank deficiency. Systematic uncertainties may formally restore invertibility,
13 but the resulting near-singular directions yield unphysically inflated χ^2 values in global goodness-
14 of-fit tests. We present the range-projected χ^2 , a test statistic that restricts the goodness-of-fit
15 test to the subspace carrying independent statistical information, yielding a χ^2 with $N_{\text{bins}} - N_{\text{null}}$
16 degrees of freedom, where N_{null} is the number of independent constraints. We derive N_{null} from
17 the bin-combination structure of the measurement, and show how it decomposes into structural
18 contributions determined a priori from the binning geometry and kinematic contributions that
19 depend on the phase-space occupancy. The method is validated with an analytical toy model and a
20 realistic simulated neutrino–argon cross-section measurement including unfolding and multi-source
21 systematic uncertainties.

22 **KEYWORDS:** Analysis and statistical methods

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39 1 Introduction

40 Modern neutrino cross-section literature is experiencing an increase of published measurements
41 reporting multiple differential distributions extracted from the same event sample. MicroBooNE
42 has reported simultaneous measurements of cross sections for final states with and without protons
43 [1, 2], multi-differential QE-like cross section measurements in kinematic imbalance variables
44 [3, 4], and triple-differential inclusive results [5]. Other relevant examples include the multiple
45 final-state kinematic projections from T2K [6, 7], and double-differential inclusive results from
46 NOvA [8–10]. In each of these analyses, the same selected events populate bins in multiple
47 reported distributions, leading to strong inter-measurement statistical correlations. Separate but
48 related efforts have produced simultaneous measurements across different nuclear targets [11–13]
49 and combined neutrino-antineutrino results [14], where correlations arise from shared systematic
50 uncertainties rather than shared events.

51 When multiple distributions are extracted from a common event sample, the event-sharing
52 structure imposes exact linear constraints among the bin counts. If every selected event populates
53 one bin in each distribution, then the total number of events in all of them is exactly identical. More
54 generally, the pattern of shared bin occupancy across distributions determines a set of constraints that
55 reduce the effective dimensionality of the measurement below the total bin count. These constraints
56 are a property of the measurement itself, independent of any particular statistical treatment.

The blockwise unfolding framework of Ref. [15] provides a rigorous method for computing the full cross-distribution covariance matrices in such measurements. In this approach, all bins corresponding to the same measured observable in the multi-distribution analysis are organized in blocks, each corresponding to a single differential cross-section distribution. Each bin block is unfolded independently using any standard method (e.g. iterative D’Agostini unfolding [16] or Wiener-SVD [17]), while the full covariance matrix, encoding the correlations between bins from different distributions, is constructed and propagated through the cross section extraction procedure. The statistical covariance for bins in different blocks is determined by the number of events they share. The constraints described above can manifest as a rank deficiency in the statistical covariance matrix, as certain directions in bin space carry identically zero variance. The number of these degenerate directions is exactly predictable from the binning geometry.

A natural use of these combined measurements is computing global goodness-of-fit metrics, such as a χ^2 statistic, that test a model against all distributions simultaneously. Compared to evaluating each distribution independently, a global test that accounts for all inter-block correlations provides the maximum discriminating power available from the data. Such global metrics are essential for interaction model tuning efforts [18–20], where measurements of multiple kinematic distributions provide complementary sensitivity to different aspects of the underlying physics. However, a global χ^2 that does not account for the constrained directions yields a test with the wrong effective number of degrees of freedom. When the statistical covariance matrix is constructed under the Poisson assumption, the constraints appear as exact null directions that prevent direct matrix inversion. Systematic uncertainties may formally restore invertibility, but the resulting near-singular directions produce unphysically inflated χ^2 values. In either case, the test is unreliable unless the constrained directions are properly identified and excluded.

We present a method to correctly compute a global goodness-of-fit test statistic for multi-distribution measurements with shared events. The approach is based on identifying the constrained directions from the bin structure of the measurement and projecting the statistic onto the subspace where independent statistical information is available. We refer to the resulting test statistic as the range-projected χ^2 . We demonstrate the method using a pedagogical toy example and connect it to the established statistical literature on hypothesis testing with singular covariance matrices.

2 Constraints from shared events

Consider a neutrino cross-section analysis simultaneously measuring B different flux-integrated differential cross sections, each with N_b true bins. If the measurement is extracted from a common event sample then each selected event will populate exactly one bin in each distribution it belongs to. Recent MicroBooNE analyses [21] have adopted the blockwise unfolding approach presented in Ref. [15] to simultaneous cross-section measurements.

This method provides a systematic framework for reporting the correlations between the different extracted differential cross sections. It provides a general prescription to construct statistical covariance matrices that respect the correlations between bins in different distributions or blocks. The full inter-block covariance matrix constructed in reconstructed space is propagated through unfolding using a block-diagonal unfolding matrix, obtained as a direct sum of the individual unfolding matrices computed for each block. The result of the combined measurement is a vector of

98 $N_{\text{bins}} = \sum_{b=1}^B N_b$ cross-section values with an $N_{\text{bins}} \times N_{\text{bins}}$ covariance matrix. The total covariance
 99 decomposes as $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$, where the statistical covariance component, computed from the
 100 shared-event structure, encodes the constraints described in the introduction.

101 2.1 Statistical covariance from shared events

102 The core idea behind the treatment of the statistical correlations in the blockwise unfolding approach
 103 is the partition of any arbitrary pair of overlapping kinematic bins X and Y into a set of three non-
 104 overlapping bins u , v , and w . The events in the intersection $X \cap Y$ are assigned to bin v , while bin
 105 u contains the events in the relative complement of Y in X (i.e. $X \setminus Y$), and bin w corresponds to
 106 the relative complement of X in Y (i.e. $Y \setminus X$). Denoting the number of predicted events in each
 107 bin as n_i , and assuming these follow independent Poisson distributions, the covariance between the
 108 counts in bins X and Y is given by [15]:

$$\begin{aligned} \text{Cov}(n_X, n_Y) &= \text{Cov}(n_u + n_v, n_v + n_w) \\ &= \text{Cov}(n_u, n_v) + \text{Cov}(n_u, n_w) + \text{Cov}(n_v, n_v) + \text{Cov}(n_v, n_w) \\ &= \text{Var}(n_v), \end{aligned} \quad (2.1)$$

109 as by definition bins u , v , and w have no events in common. This implies that the covariance
 110 between any two bins can be estimated from the variance of the events they share.

111 The variance for a set of Monte Carlo (MC) events can be estimated from the sum of their
 112 squared weights. This gives us a general prescription to calculate the elements of the statistical
 113 covariance matrix as:

$$(C_{\text{stat}})_{ij} = \sum_{e \in (i \cap j)} w_e^2, \quad (2.2)$$

114 where w_e is the weight associated to event e belonging to the intersection of bins i and j . In the
 115 case of measured data the weights are 1, and the covariance element simplifies to the number of
 116 events shared between the bins n_{ij} .

117 After propagation through the block-diagonal unfolding, the statistical covariance matrix in
 118 unfolded space inherits the same rank structure, as the unfolding matrices for each individual block
 119 are invertible. This rank structure has a direct interpretation in terms of the number of distinct
 120 event-sharing patterns across distributions, which we discuss next.

121 2.2 Combination matrix and null space

122 The previous discussion shows that the statistical covariance between any two bins is determined
 123 solely by the events they share. We now formalize this observation to derive an exact formula for
 124 the rank of C_{stat} , and therefore for the number of directions in bin space that carry zero statistical
 125 variance.

126 For each event e , we can define the indicator vector $\phi_e := (\phi_{e,i})_{i=1}^{N_{\text{bins}}}$ such that $\phi_{e,i} = 1$ if
 127 $i \in \text{bins}(e)$ and $\phi_{e,i} = 0$ if $i \notin \text{bins}(e)$, where $\text{bins}(e)$ is the set of bins event e populates across all
 128 distributions. The indicator vector acts as an identifier of each event in the combined measurement.
 129 For example, in a two-block measurement with 3 bins each, an event populating bin 2 of block 1
 130 and bin 1 of block 2 has indicator vector $\phi_e = (0, 1, 0, 1, 0, 0)$. The statistical covariance matrix

131 can then be re-written in terms of indicator vectors as:

$$C_{\text{stat}} = \sum_e w_e^2 \phi_e \phi_e^\top. \quad (2.3)$$

132 This outer-product decomposition makes the rank structure of C_{stat} clear. The rank of the
 133 statistical covariance matrix equals the dimension of $\text{span}\{\phi_e\}$, which depends only on the distinct
 134 indicator vectors present, not on their weights. Since all weights are $w_e > 0$ no indicator vector is
 135 removed from the sum, and the column space of C_{stat} is exactly $\text{span}\{\phi_e\}$.

136 We define the bin-combination matrix Ω as the matrix whose rows are the unique indicator
 137 vectors among $\{\phi_e\}$. Each row of Ω represent a distinct pattern of bin occupancy across the blocks.
 138 By construction the row space of Ω equals $\text{span}\{\phi_e\}$, and therefore:

$$\text{rank}(C_{\text{stat}}) = \text{rank}(\Omega). \quad (2.4)$$

139 The number of degenerate directions is then:

$$N_{\text{null}} = N_{\text{bins}} - \text{rank}(\Omega). \quad (2.5)$$

140 Physically, each of the N_{null} directions corresponds to a linear combination of bins along which
 141 the measurement has identically zero variance. Any component of the residual vector along such
 142 a direction is not perceived by the statistical covariance; it is a direction that the measurement has
 143 no statistical power to constrain. Importantly, both Ω and N_{null} are directly computable from the
 144 MC event-to-bin mapping alone, before constructing any covariance matrix, confirming that the
 145 constraints are a property of the measurement geometry rather than the statistical treatment.

146 A vector $\mathbf{x}$ is in the null space of C_{stat} if and only if, for every event e :

$$\sum_{i \in \text{bins}(e)} x_i = 0, \quad (2.6)$$

147 or equivalently $\Omega \mathbf{x} = 0$. Each unique indicator vector acts as a linear constraint on the null-space
 148 vectors, requiring that the components of $\mathbf{x}$ sum to zero over the bins populated by the different
 149 events. The null space directions can be divided into two physically distinct categories: structural
 150 constraints that follow from the block topology alone, and kinematic constraints that depend on
 151 which regions of phase space are populated.

152 2.3 Determination of the structural null space

153 The combination matrix Ω presented previously is constructed from MC events. One determines
 154 which events populate the different bins and collects the unique indicator vectors. However, the
 155 indicator vectors that can appear are constrained by two distinct mechanisms. The first is purely
 156 structural, the geometry of the binning scheme forbids certain bin combinations. The second is
 157 kinematic, correlations between the different variables render some combinations unreachable even
 158 though they are geometrically allowed. These constraints are imposed by the underlying physics
 159 effects, mainly the differential cross section and the detector acceptance. In the following, we show
 160 that the structural constraints can be determined a priori, without generating MC events, from the
 161 bin edges used.

162 The variables used in a measurement are drawn from a set $\{v_1, \dots, v_D\}$. Each block b uses
 163 a subset of these variables, $\text{vars}(b) \subseteq \{v_1, \dots, v_D\}$. Restricting the discussion to combinations of
 164 single- and double-differential measurements, we have $|\text{vars}(b)| \in \{1, 2\}$ for every b . For each
 165 variable v used in a given block b , the binning scheme defines a set of bin edges $E_v^{(b)}$ that partition
 166 the range of v into intervals. In the case of a two-dimensional block, the edges in the binning
 167 variable may be slice-dependent, but nevertheless these can be collected into a single set of edges
 168 per variable across slices. For each variable $v \in \bigcup_b \text{vars}(b)$, we can define the refined edges:

$$\mathcal{E}_v = \bigcup_{b: v \in \text{vars}(b)} E_v^{(b)}, \quad (2.7)$$

169 with $M_v = |\mathcal{E}_v| - 1$ denoting the number of refined bins for variable v . Since $\mathcal{E}_v$ contains all the
 170 bin edges from every block that uses that variable, each refined bin is contained entirely within one
 171 original bin of every such block.

172 The common refinement of all blocks is the Cartesian product of the refined bins across all
 173 variables, denoted by $\mathcal{R}$. Each cell $c \in \mathcal{R}$ specifies a refined bin index for every variable in the
 174 measurement. For every block b , there exists a unique mapping $\beta_b(c)$ between each cell c and an
 175 original bin in said block. The elements of the indicator vector $\phi_c := (\phi_{c,i})_{i=1}^{N_{\text{bins}}}$ associated to cell
 176 c can be written as:

$$\phi_{c,i} = \begin{cases} 1 & \text{if } i = \beta_b(c) \text{ for any block } b, \\ 0 & \text{otherwise.} \end{cases} \quad (2.8)$$

177 If block b does not use variable v then β_b is independent of the refined bin index in v . Therefore,
 178 multiple cells may produce identical indicator vectors.

179 The structural combination matrix Ω_{struct} is defined as the matrix whose rows are the unique
 180 indicator vectors from the cells of the common refinement $\{\phi_c : c \in \mathcal{R}\}$. By construction, every
 181 indicator vector that appears in the Ω matrix obtained from the MC events is also present in Ω_{struct} ,
 182 therefore:

$$\text{rank}(\Omega) \leq \text{rank}(\Omega_{\text{struct}}) \leq N_{\text{bins}}. \quad (2.9)$$

183 This hierarchy motivates a decomposition of the total number of degenerate directions:

$$\begin{aligned} N_{\text{null}} &= N_{\text{bins}} - \text{rank}(\Omega_{\text{struct}}) + \text{rank}(\Omega_{\text{struct}}) - \text{rank}(\Omega) \\ &= N_{\text{null}}^{\text{struct}} + N_{\text{null}}^{\text{kinem}}. \end{aligned} \quad (2.10)$$

184 The first term, $N_{\text{null}}^{\text{struct}}$, counts null directions that arise purely from the binning geometry and is
 185 computable a priori from the bin edges alone. The second term, $N_{\text{null}}^{\text{kinem}}$, accounts for additional null
 186 directions introduced by phase-space restrictions that exclude geometrically allowed combinations
 187 and can only be determined from Monte Carlo. A description of the algorithm used to compute
 188 Ω_{struct} , including an optimization that restricts the enumeration to shared variables, is presented in
 189 Appendix A.

190 The presence of kinematic null directions indicate that certain regions of the combined bin
 191 space are empty, which may signal that the binning extends beyond the kinematic reach of the
 192 measurement or that it is finer than the available statistics can support. In such cases, merging or
 193 removing the offending bins may be preferable to relying on a projection to handle the resulting

constraints. The structural null directions, by contrast are an unavoidable consequence of measuring multiple distributions from shared events and cannot be eliminated by rebinning.

The structure of the null space of Ω_{struct} depends on how the bin edges for the different blocks overlap. In the simplest case, when blocks do not share variables, every combination of one bin per block is allowed. The resulting null space is $(B - 1)$ -dimensional, with a single constraint that requires all block totals to be equal. When blocks share one or more variables, the allowed combinations are restricted and the dimension of the null space can be different from $B - 1$. The number of null directions depends on how shared variable bin edges are aligned between blocks. Aligned slice edges decouple the measurement into different sub-problems, adding additional sum rules and increasing $N_{\text{null}}^{\text{struct}}$. On the other hand, misaligned edges couple adjacent distributions, potentially reducing the number of constraints. In the general case, the interplay between shared variables, bin edge alignment, and block topology makes a closed-form expression impractical, but the computation of $\text{rank}(\Omega_{\text{struct}})$ from the bin edge specifications is straightforward.

The structural and kinematic components of the null space respond differently to systematic uncertainties. Whether these cause the rank deficiency of C_{stat} to persist, partially resolve, or fully resolve in the total covariance C_{total} depends on the nature of the systematic treatment, which we discuss next.

2.4 Effect of systematic uncertainties

Previously, we established that the event-sharing structure imposes $N_{\text{null}} = N_{\text{bins}} - \text{rank}(\Omega)$ exact linear constraints on the bin counts, which manifest as a rank deficiency of C_{stat} . Whether the total covariance matrix inherits the rank deficiency depends on which of these null directions receive support from the systematic uncertainties. Now, we examine the mechanisms by which different systematic treatments do or do not populate the null space of C_{stat} .

A systematic variation s produces a modified vector of expected events $\mathbf{n}_s$, whose departure from the central-value prediction can be expressed in terms of a shift vector:

$$\delta_s = \mathbf{n}_s - \mathbf{n}_{\text{CV}}. \quad (2.11)$$

The different contributions to C_{syst} are constructed from the outer product of these shift vectors. A null direction $\mathbf{x}$ of C_{stat} remains a null direction of C_{total} if and only if $\mathbf{x} \cdot \delta_s = 0$ for every systematic variation in the uncertainty budget. The question is therefore whether the shift vectors have nonzero projections onto the structural and kinematic null spaces.

Consider first the simplest case, where a systematic s acts by reweighting the reconstructed event counts directly. Each event e is assigned a modified weight $w_e^{(s)}$ while preserving its bin assignments. The shift vector can then be written as:

$$\delta_s = \sum_e (w_e^{(s)} - w_e) \phi_e. \quad (2.12)$$

For any null vector $\mathbf{x}$ which satisfies the null-space condition of Eq. (2.6), the projection of $\mathbf{x}$ onto

the shift vector gives:

$$\begin{aligned}
\mathbf{x} \cdot \boldsymbol{\delta}_s &= \sum_e (w_e^{(s)} - w_e) (\boldsymbol{\phi}_e \cdot \mathbf{x}) \\
&= \sum_e (w_e^{(s)} - w_e) \sum_{i=1}^{N_{\text{bins}}} \phi_{e,i} x_i \\
&= 0,
\end{aligned} \tag{2.13}$$

since the inner sum vanishes for every event. Direct reweighting therefore preserves the entire null space, adding no rank to C_{total} . Flux uncertainties, which modify the neutrino energy spectrum through per-event weights, are the classic example. Cross-section model variations and hadronic reinteraction uncertainties, when implemented as per-event reweighting, behave identically.

A qualitatively different situation arises when the systematic uncertainties are propagated through the response matrix rather than changing the event counts per bin directly. In this approach, adopted as the standard prescription in MicroBooNE for neutrino interaction uncertainties [15], a modified response matrix Δ'_s , mapping true bins to reconstructed bins under the given variation, is constructed for each systematic uncertainty. This matrix is obtained either by reweighting the events used to build the nominal response matrix or from an independent MC sample with varied simulation parameters. In this case, the predicted reconstructed counts for systematic variation s are:

$$\mathbf{n}_s = \Delta'_s \boldsymbol{\varphi}_{\text{CV}}, \tag{2.14}$$

where $\boldsymbol{\varphi}_{\text{CV}}$ is the central-value truth prediction. The shift vector is then:

$$\boldsymbol{\delta}_s = (\Delta'_s - \Delta) \boldsymbol{\varphi}_{\text{CV}}. \tag{2.15}$$

Since Δ is block-diagonal, the difference $\Delta'_s - \Delta$ acts independently on each block, projecting the nominal truth vector through block-specific changes in efficiency and migration. The per-event coupling that makes direct reweighting respect all null directions is absent in this prescription. When the weight of an event changes, it affects simultaneously every block it populates; the response-matrix method, in contrast, processes each block independently through its own sub-matrix. The projection of $\boldsymbol{\delta}_s$ onto any null vector, whether structural or kinematic, therefore involves differences in how each block responds to the same truth-level prediction, and in general is nonzero. Response-matrix variations thus break both types of null directions, regardless of how Δ'_s is obtained.

A third mechanism applies when systematic uncertainties are evaluated by directly comparing reconstructed event counts from independent MC samples simulated with nominal and varied parameters. In this case, the shift vector takes the form:

$$\boldsymbol{\delta}_s = \sum_{e'} w_{e'}^{(s)} \boldsymbol{\phi}_{e'}^{(s)} - \sum_e w_e \boldsymbol{\phi}_e, \tag{2.16}$$

where the first sum runs over the events in the varied-model sample and the second sum over the nominal sample. Since each event in both samples populates exactly one bin per block, their difference projects to zero onto the structural null directions by an analogous argument to that used for the direct reweighting. Kinematic null directions, however, can be lifted if the variation scatters events into reconstructed-bin combinations absent in the nominal sample. In that

case, new indicator vectors appear that are not rows of Ω , potentially breaking the corresponding constraints. Detector systematic uncertainties, which usually require independent samples with modified simulation parameters that cannot be captured by event reweighting, are the typical example where this prescription is applied.

In practice, whether the eigenvalues of C_{total} along the formerly null directions are exactly zero or merely smaller than the physical eigenvalues, including these directions in a χ^2 test via standard matrix inversion produces inflated values with the wrong effective number of degrees of freedom. A valid goodness-of-fit test must therefore be restricted to the subspace where the measurement carries genuine sensitivity, which is the role of the projection method developed in the following section.

3 Range-projected χ^2 test statistic

The discussion presented in Sec. 2 established that the event-sharing structure of the multi-distribution measurement imposes N_{null} linear constraints, encoded in the null space of the combination matrix Ω , which emerges as a rank deficiency in C_{stat} . A global goodness-of-fit test statistic must therefore be formulated in the subspace orthogonal to these constrained directions, where independent statistical information is available.

The constraints are most naturally identified in reconstructed space, where the combination matrix is constructed. The non-null subspace can be obtained decomposing $\Omega^T \Omega$. This matrix is a square $N_{\text{bins}} \times N_{\text{bins}}$ matrix with the same column space and null space as Ω itself, making it a more compact representation of the same information. We can write its eigendecomposition as:

$$\Omega^T \Omega = V S V^T, \quad (3.1)$$

where $S = \text{diag}(s_1, \dots, s_{N_{\text{bins}}})$ contains the eigenvalues in decreasing order and V is the orthogonal matrix of eigenvectors. Of these eigenvalues, $r = \text{rank}(\Omega)$ are nonzero. Partitioning $V = [V_r \mid \bar{V}_r]$ into the columns associated with the nonzero and zero eigenvalues, the columns of V_r span the subspace of reconstructed bin space that is not constrained by the event-sharing structure. The orthogonal projector onto this subspace is:

$$P_{\text{reco}} = V_r V_r^T. \quad (3.2)$$

Note that P_{reco} is identical to the projector onto the range of C_{stat} as defined in Eq. (2.3), since C_{stat} and $\Omega^T \Omega$ share the same column space by construction. The combination matrix formulation makes explicit the fact that the projector is determined by the measurement geometry.

Cross-section measurements are typically reported in unfolded space, and it is in that space where goodness-of-fit tests are evaluated. The blockwise unfolding matrix U , which is block-diagonal and invertible, maps reconstructed-space vectors onto unfolded space. The non-null basis vectors propagate as $W_r = U V_r$. Since U is not orthogonal in general, the columns of W_r are not orthonormal, and the projector onto their span takes the general form:

$$P_{\text{true}} = W_r (W_r^T W_r)^{-1} W_r^T. \quad (3.3)$$

This projector identifies the subspace of unfolded bin space that carries independent statistical information, with detector response and unfolding effects properly accounted for. Since U is

invertible, W_r has the same rank as V_r and the number of constrained directions is preserved. A test statistic constructed from the projected covariance matrix would have $r = N_{\text{bins}} - N_{\text{null}}$ effective degrees of freedom.

In practice, the eigenvalues of $\Omega^T \Omega$ that are technically zero take small numerical values. The threshold for identifying these can be set relative to the largest eigenvalue (e.g., $s_i/s_1 < 10^{-10}$). In most cases, a clear spectral gap of several orders of magnitude separates the physical eigenvalues from the numerical noise, making the identification unambiguous.

3.1 Test statistic definition

The goodness-of-fit test is evaluated in unfolded space, where the residual vector is defined as:

$$\mathbf{r} = \hat{\boldsymbol{\varphi}} - A_C \boldsymbol{\varphi}_{\text{pred}}, \quad (3.4)$$

where $\hat{\boldsymbol{\varphi}} = U\mathbf{d}$ is the unfolded background-subtracted data, $A_C = U\Delta$ is the additional smearing matrix [17], and $\boldsymbol{\varphi}_{\text{pred}}$ is a truth-level model prediction. The range-projected χ^2 in unfolded space is:

$$\chi_{\text{proj}}^2 = \mathbf{r}^T (P_{\text{true}} V_{\text{total}} P_{\text{true}})^+ \mathbf{r}, \quad (3.5)$$

where $V_{\text{total}} = U C_{\text{total}} U^T$ is the total covariance matrix in unfolded space and $^+$ denotes the Moore–Penrose pseudoinverse.

The pseudoinverse can be avoided entirely by reducing the problem to r dimensions. One can verify that the pseudoinverse of $P_{\text{true}} V_{\text{total}} P_{\text{true}}$ takes the closed form:

$$(P_{\text{true}} V_{\text{total}} P_{\text{true}})^+ = W_r (W_r^T V_{\text{total}} W_r)^{-1} W_r^T. \quad (3.6)$$

Substituting into Eq. (3.5) and defining the reduced residual $\tilde{\mathbf{r}} = W_r^T \mathbf{r}$ and the reduced covariance matrix $\tilde{V}_{\text{total}} = W_r^T V_{\text{total}} W_r$, the test statistic becomes:

$$\chi_{\text{proj}}^2 = \tilde{\mathbf{r}}^T \tilde{V}_{\text{total}}^{-1} \tilde{\mathbf{r}}, \quad (3.7)$$

which involves only a standard inversion of the $r \times r$ matrix $\tilde{V}_{\text{total}}$. Under the null hypothesis, χ_{proj}^2 follows a χ^2 distribution with $r = N_{\text{bins}} - N_{\text{null}}$ degrees of freedom.

An identical construction can be carried out entirely in reconstructed space. Defining the reconstructed-space residual as $\boldsymbol{\delta} = \mathbf{d} - \Delta \boldsymbol{\varphi}_{\text{pred}}$, the corresponding projected χ^2 is given by:

$$\begin{aligned} \chi_{\text{reco}}^2 &= \boldsymbol{\delta}^T (P_{\text{reco}} C_{\text{total}} P_{\text{reco}})^+ \boldsymbol{\delta} \\ &= \boldsymbol{\delta}^T V_r (V_r^T C_{\text{total}} V_r)^{-1} V_r^T \boldsymbol{\delta} \\ &= \tilde{\boldsymbol{\delta}}^T \tilde{C}_{\text{total}}^{-1} \tilde{\boldsymbol{\delta}}, \end{aligned} \quad (3.8)$$

where the reduced form follows directly from the orthonormality of V_r . Without projectors, the two quadratic forms are equivalent, as the unfolding matrix cancels in the unfolded space calculation. This is a well-known result when using the additional smearing matrix to compare model predictions to unfolded data. However, using projectors breaks this equivalence. In this case, the test statistic in unfolded space contains factors of $U^T U$, as can be seen by expressing it in terms of reconstructed-space quantities:

$$\chi_{\text{proj}}^2 = \boldsymbol{\delta}^T U^T U V_r (V_r^T U^T U C_{\text{total}} U^T U V_r)^{-1} V_r^T U^T U \boldsymbol{\delta}, \quad (3.9)$$

which reduces to Eq. (3.8) only when $U^\top U = I$, i.e. when the unfolding matrix is orthogonal. Since unfolding matrices are generically not orthogonal, the two test statistics yield different numerical values for the same data and prediction. The unfolded-space formulation is the appropriate one for testing published cross-section results, as it accounts for effects introduced by the unfolding in the definition of the non-null subspace.

3.2 Statistical validity

Under the null hypothesis, the residual $\mathbf{r}$ is drawn from a multivariate Gaussian distribution with mean zero and covariance V_{total} . The reduced residual $\tilde{\mathbf{r}} = W_r^\top \mathbf{r}$ is a linear transformation of a Gaussian vector, therefore Gaussian itself with mean zero and covariance $\tilde{V}_{\text{total}} = W_r^\top V_{\text{total}} W_r$. Provided that $\tilde{V}_{\text{total}}$ is positive definite, guarantees that $\tilde{\mathbf{r}}^\top \tilde{V}_{\text{total}}^{-1} \tilde{\mathbf{r}} \sim \chi^2(r)$. The positive definiteness of $\tilde{V}_{\text{total}}$ requires V_{total} to have non-zero variance along every direction in the range of the projector P_{true} , which is generically satisfied. This can be verified by checking that all eigenvalues of $\tilde{V}_{\text{total}}$ are strictly positive.

The validity of the range-projected χ^2 can also be understood from the general theory of quadratic forms of correlated normal variables. Theorem 9.2.2 of Ref. [22] establishes that if $\mathbf{Y} \sim \mathcal{N}(0, \Sigma)$ with $\text{rank}(\Sigma) = k$, then $\mathbf{Y}^\top \Sigma^+ \mathbf{Y} \sim \chi^2(k)$ for any choice of generalized inverse Σ^+ . Assuming the null hypothesis, the projected residual $P_{\text{true}} \mathbf{r}$ is normally distributed with covariance $P_{\text{true}} V_{\text{total}} P_{\text{true}}$, which has rank r by construction. The formulation of Eq. (3.5) with the Moore–Penrose pseudoinverse is a specific instance of this general result. The key distinction from purely numerical approaches to handling singular covariance matrices is that the rank is not determined by an eigenvalue threshold applied to the covariance, but is predicted exactly via the combination matrix Ω from the structure of the measurement.

3.3 Relation to alternative approaches

4 Toy example

The formalism developed previously is now illustrated with a toy model that captures the essential features of a multi-distribution neutrino cross-section measurement: a two-dimensional underlying physics process, a simple detector model, and a multi-block binning scheme in which events are shared across distributions. The toy is design to be minimal but non-trivial. A three-block structure produces a rank-deficient statistical covariance matrix, and a Feldman–Cousins study demonstrates the consequences of ignoring or correctly treating the resulting null directions.

4.1 Model and measurement setup

The toy model mimics a muon neutrino charged-current quasi-elastic-like interaction, producing a muon characterized by its momentum p_μ and scattering angle θ_μ . The joint truth-level differential cross section is parametrized as:

$$f(p_\mu, \cos \theta_\mu) = p_\mu^k e^{-k p_\mu / p_0} e^{\kappa \cos \theta_\mu} (1 + \alpha p_\mu \cos \theta_\mu), \quad (4.1)$$

where the four parameters (k, p_0, κ, α) control the shape of the momentum spectrum, the level of forward peaking, and the strength of the momentum–angle correlation, respectively. A central-value (CV) model with $(k, p_0, \kappa, \alpha) = (3.0, 0.2, 2.0, 0.3)$ is used to build the response matrix and

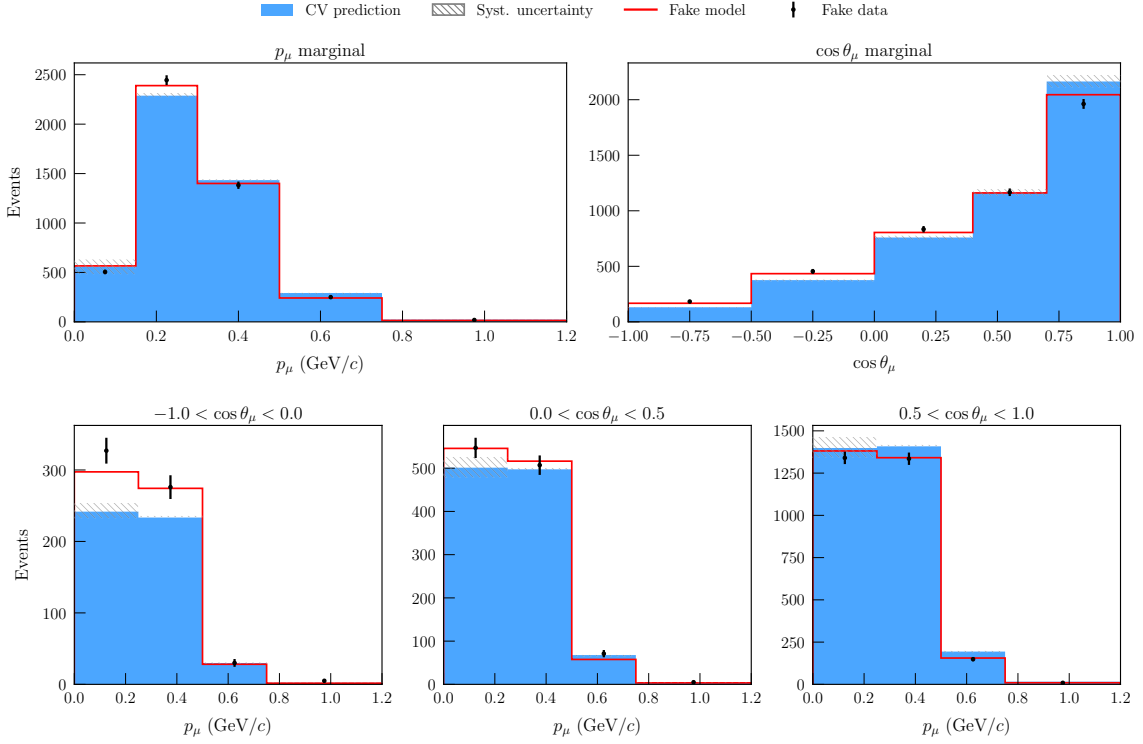


Figure 1. Predicted event distributions in the toy analysis blocks for the CV model (filled histograms) and the fake model (solid lines), together with a pseudo-data realization (points with statistical error bars). The shaded bands indicate the systematic uncertainty on the CV prediction. The top row shows the p_μ and $\cos \theta_\mu$ marginals; the bottom row shows the three $\cos \theta_\mu$ slices of the 2D block, each binned in p_μ .

estimate the covariance, while a “fake” model with $(k, p_0, \kappa, \alpha) = (3.3, 0.2, 1.8, 0.35)$ serves as the true underlying distribution from which pseudo-data are drawn. Truth events are generated via rejection sampling over $p_\mu \in [0.0, 1.2]$ GeV/c and $\cos \theta_\mu \in [-1.0, 1.0]$.

A simple detector model introduces two different effects. Momentum is smeared multiplicatively, $p_{\text{reco}} = p_{\text{true}}(1 + \delta_p)$ with $\delta_p \sim \mathcal{N}(0, \sigma_p)$ and $\sigma_p = 0.05$, while the angle is smeared additively, $\theta_{\text{reco}} = \theta_{\text{true}} + \delta_\theta$ with $\delta_\theta \sim \mathcal{N}(0, \sigma_\theta)$ and $\sigma_\theta = 1^\circ$. A sigmoid efficiency function $\varepsilon(p_{\text{reco}}) = [1 + e^{-(p_{\text{reco}} - p_{\text{thr}})/w}]^{-1}$ with threshold $p_{\text{thr}} = 0.10$ GeV/c and width $w = 0.02$ GeV/c models the turn-on of event selection at low momentum.

The analysis uses three bin blocks in reconstructed space: a p_μ marginal (5 bins), a $\cos \theta_\mu$ marginal (5 bins), and a two-dimensional distribution sliced in $\cos \theta_\mu$ (3 slices, each with 4 bins in p_μ), for a total of $N_{\text{bins}} = 22$. The bin edges of the 2D block do not align exactly with those of the 1D marginals, which is the usual situation in real analyses. A CV sample of $N_{\text{MC}} = 500,000$ truth events is generated, processed through the detector model, and used to construct the response matrix Δ . Pseudo-data samples are generated by drawing $N_{\text{data}} = 5,000$ events from the fake model.

Systematic uncertainties are estimated following the response-matrix approach described in Sec. 2.4. Interaction-model variations are obtained by reweighting the CV sample to six alternative model configurations, each producing a varied response matrix Δ_s . The shift vector $\delta_s = (\Delta_s - \Delta) \varphi$ captures the effect on the predicted reconstructed-space distribution. Detector variations are

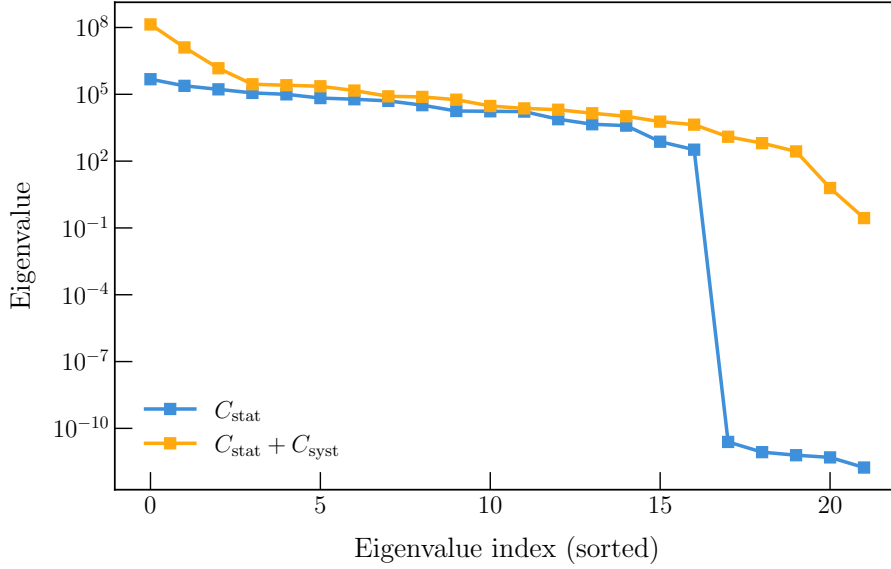


Figure 2. Expected number of signal events for reconstructed bins in block 11, used for the double-differential $(N_p, \cos \theta_{\mu l_4})$ measurement. Both the central-value MicroBooNE tune prediction (black line) and the NuWro alternative prediction (blue line) are shown. Error bars include only the MC statistical uncertainty.

obtained by generating eight independent samples with perturbed detector parameters ($\sigma_p, \sigma_\theta, p_{\text{thr}},$ and w), each producing a new response matrix from which a shift vector is computed similarly. The total systematic covariance matrix is $C_{\text{syst}} = N_{\text{var}}^{-1} \sum_i \delta_i \delta_i^T$, scaled to data statistics. The statistical covariance is computed following the prescription from Eq. (2.2), assuming unit event weights, which captures the inter-block correlations from event sharing.

Figure 1 shows the predicted event distributions for the CV and fake models in all three blocks of the toy analysis, together with a pseudo-data sample distribution and the systematic uncertainty bands.

4.2 Null-space structure

Acknowledgments

A Algorithm

The naive counting of the common refinement $\mathcal{R}$ has cost $\prod_v M_v$, which grows exponentially with the number of kinematic variables. A typical multi-distribution measurement involves $\mathcal{O}(5)$ variables each with $\mathcal{O}(10)$ refined bins, giving $\mathcal{O}(10^5)$ cells. The key insight is that only shared variables (i.e. the ones appearing in two or more blocks) generate the cross-block constraints. Dividing the variable set into shared (V_s) and private (V_p) variables, one has to enumerate only the reduced refinement $\prod_{v \in V_s} M_v$. For each of the shared-variable cells, the bins from blocks that depend on the shared variable are determined by lookup, while blocks whose binning variable is private contribute their bins independently via a local Cartesian product. If the number of shared variables is small the computational cost is dramatically reduced.

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